

Middlebox Detection and Chaining Analysis Extension of NOMS 2024 — Pournaghshband & Reiher

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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Problem Statement — The Internet is not transparent

- Middleboxes sit silently between sender and receiver
- They shape, compress, prioritize traffic without consent
- End hosts cannot see them — no standard detection mechanism
- Real-world impact: net neutrality violations, double compression, debugging blind spots

Research Question:

Can end hosts detect middleboxes using only packet loss measurements — and what happens when multiple middleboxes are chained?

Paper's Framework — Pournaghshband & Reiher, NOMS 2024



- Send probe-A (high entropy) and probe-B (low entropy) trains
- Measure loss rate difference: $\Delta I = I_A - I_B$
- If $\Delta I > \tau = 20\%$ → middlebox detected
- Paper proves three MBs individually detectable: Shaper, Compressor, SPQ
- Paper limitation: assumes single static middlebox (Section III-A)
- Section VIII explicitly lists chaining as future work

Formula: $\Delta I = I_A - I_B > \tau$

Our Contributions — What we add beyond the paper

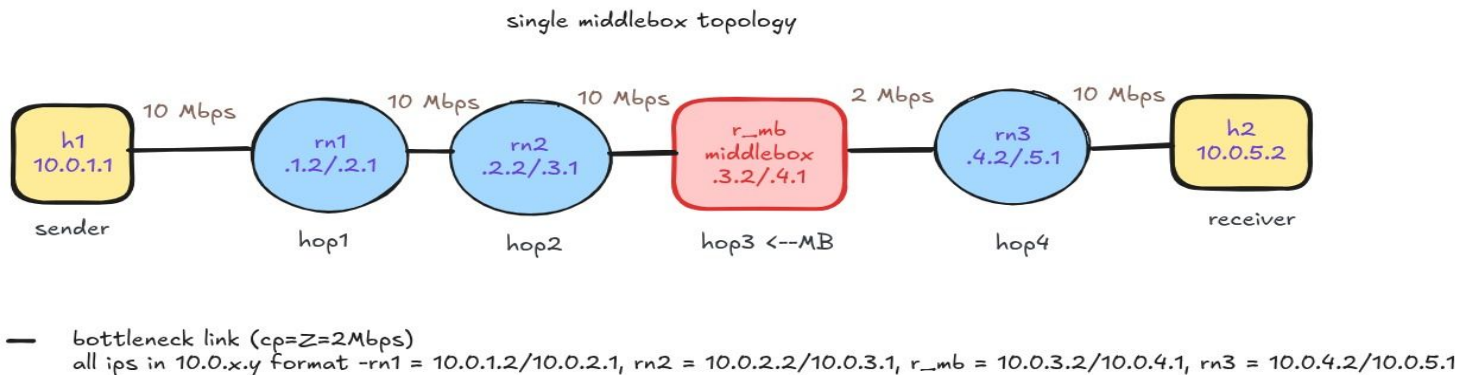
Three novel contributions:

1. **Chaining Experiments** — first empirical study of multiple chained middleboxes
2. **Masking Analysis** — quantifying how much one MB hides another
3. **TTL Localization Failure** — proving TTL probing cannot detect behavioral MBs

Testbed Topology — Mininet on Native Ubuntu

22.04

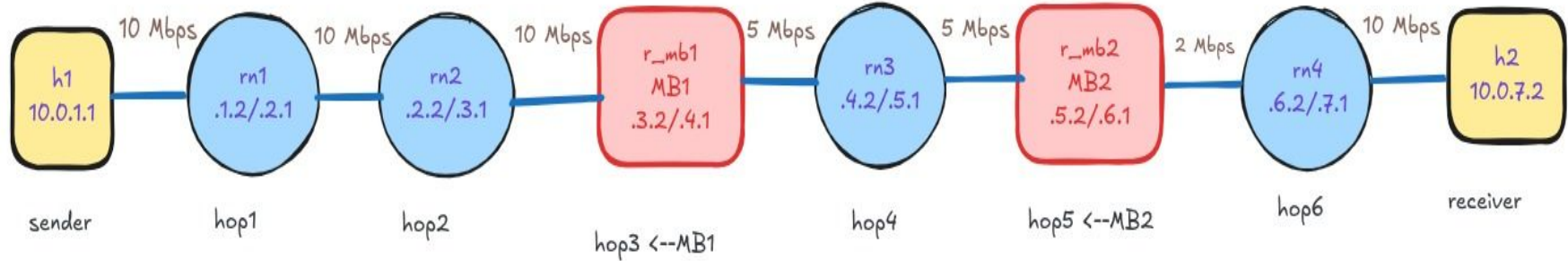
Single MB Topology:



- 2 normal routers before MB, 1 after — realistic ISP-like path
- 2 Mbps bottleneck link after r_mb — paper's $cp = Z = 2$ Mbps model
- Chain topologies extend this with $r_mb1 \rightarrow r_mb2 \rightarrow r_mb3$

Two chained Middlebox Topology

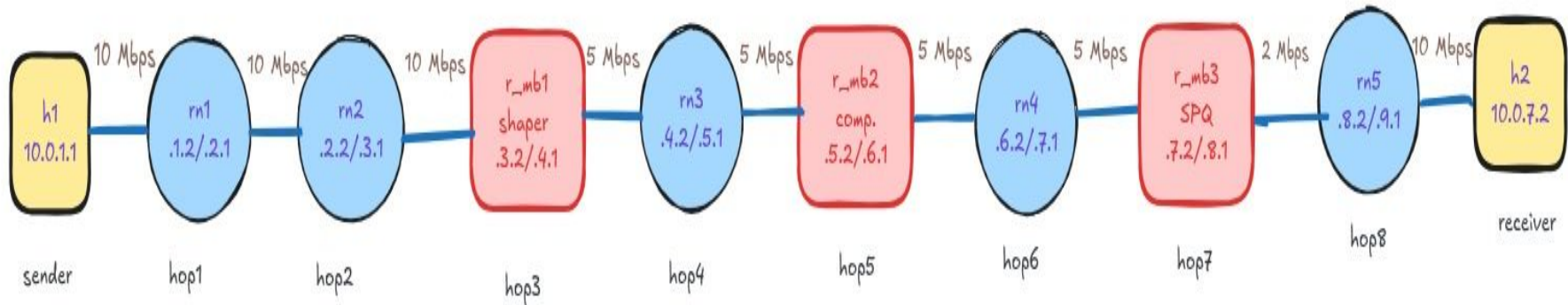
2-chain topology (EXP 4,5,6)



- EXP4: r_mb1 = shaper r_mb2 = compressor
- EXP5: r_mb1 = shaper r_mb2 = SPQ
- EXP6: r_mb1 = compressor r_mb2 = SPQ

Three chain Middlebox Topology

3-chain topology (EXP 7) - SHAPER → COMPRESSOR → SPQ



Three Middlebox Implementations — Standard Linux tc

Standard Linux tc — same as production ISP equipment

MB	Method	Classification	tc Command
Shaper	Linux tc HTB	Source port 9000	tc qdisc htb rate 2mbit
Compressor	nfqueue + zlib RFC 2394	Payload entropy (zlib ratio)	nfqueue → tc HTB
SPQ	Linux tc PRIO	DSCP EF field	tc qdisc prio + HTB

Key point: MB classifies independently — sender marks nothing for shaper and compressor

Probe Design and Detection Logic

How we probe each middlebox:

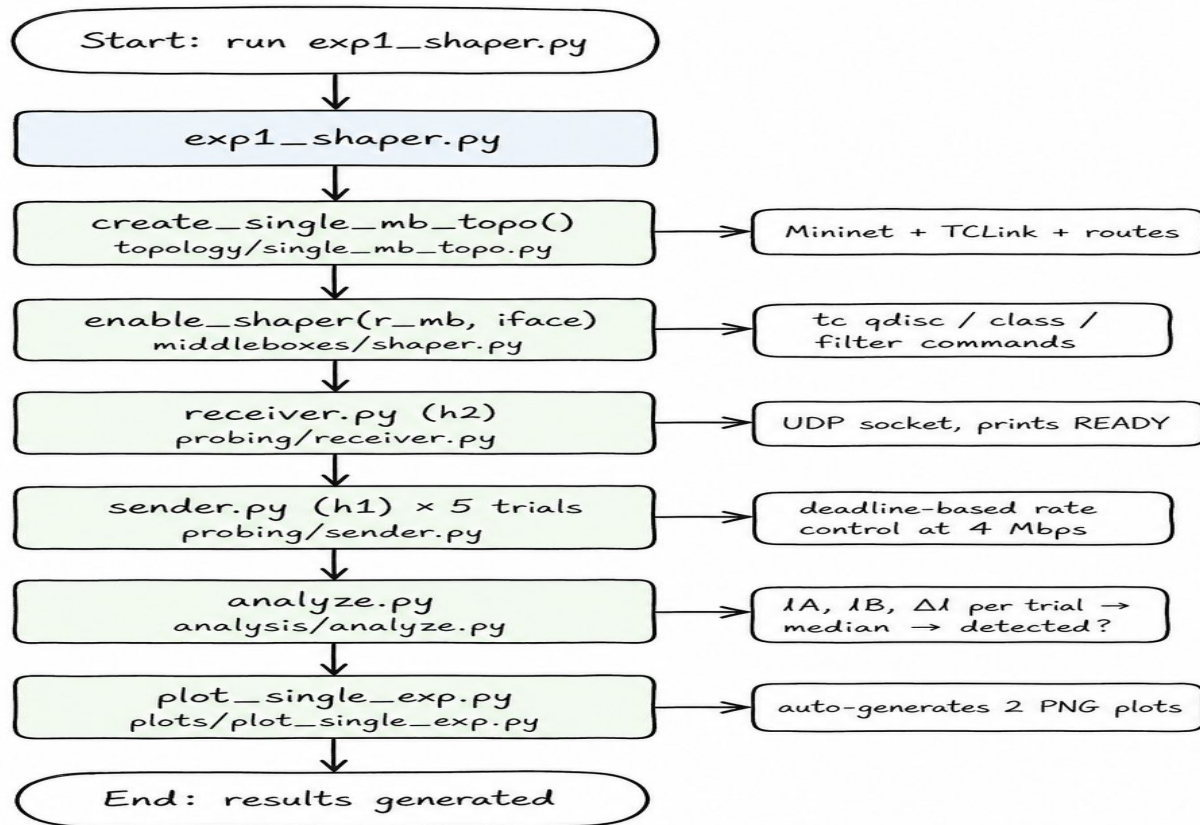
```
probe-A: [random bytes] [random bytes] ... × 1000 → 1A  
         wait 30s ( $\Lambda$ )  
probe-B: [zero bytes]    [zero bytes]    ... × 1000 → 1B
```

SPQ special case — interleaved LP/HP phases (paper Figure 3):

```
LP: [L] [H H H H] [L] [H H H H] ... → 1A (L packets starve)  
HP: [H] [L L L L] [H] [L L L L] ... → 1B (H packets served)
```

Parameters: $p=1000$, $\Lambda=30s$, $\tau=20\%$, $r=4Mbps$, $\sigma=2Mbps$, $N'=4$

Code Flow — How one experiment runs end to end



Phase 1 Results — Single Middlebox Detection

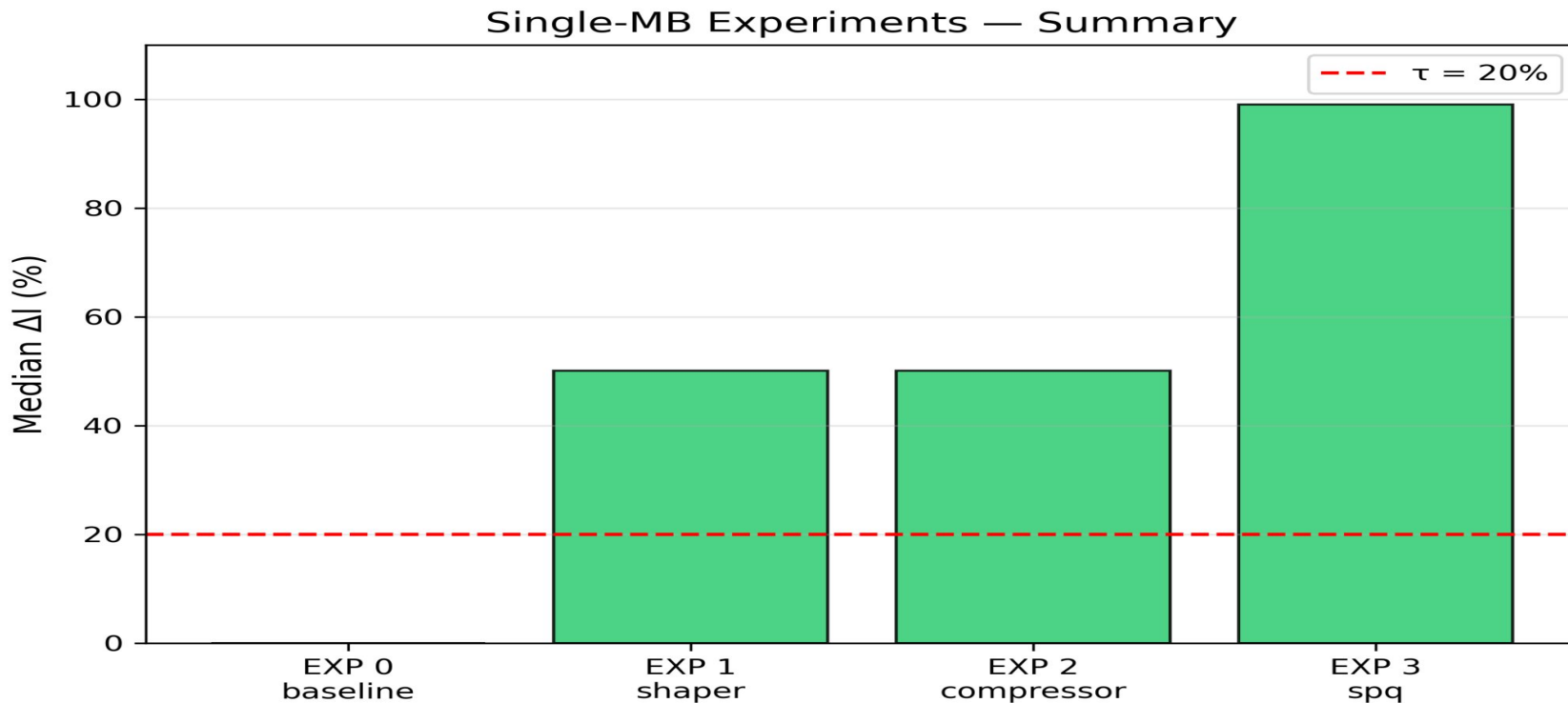
All three middleboxes detected — replicating the paper

Experiment	IA	IB	ΔI	Detected
EXP0 Baseline	50%	50%	0.0%	✗
EXP1 Shaper	50.1%	0%	50.1%	✓
EXP2 Compressor	50.1%	0%	50.1%	✓
EXP3 SPQ	99%	0%	99.0%	✓

Note: EXP0 shows $\Delta I=0\%$ with 50% absolute loss — bottleneck affects both equally, no discrimination

Phase 1 Results — Single Middlebox Detection Plot

Single MB Experiments -Summary



Phase 2: Chaining — The Masking Effect

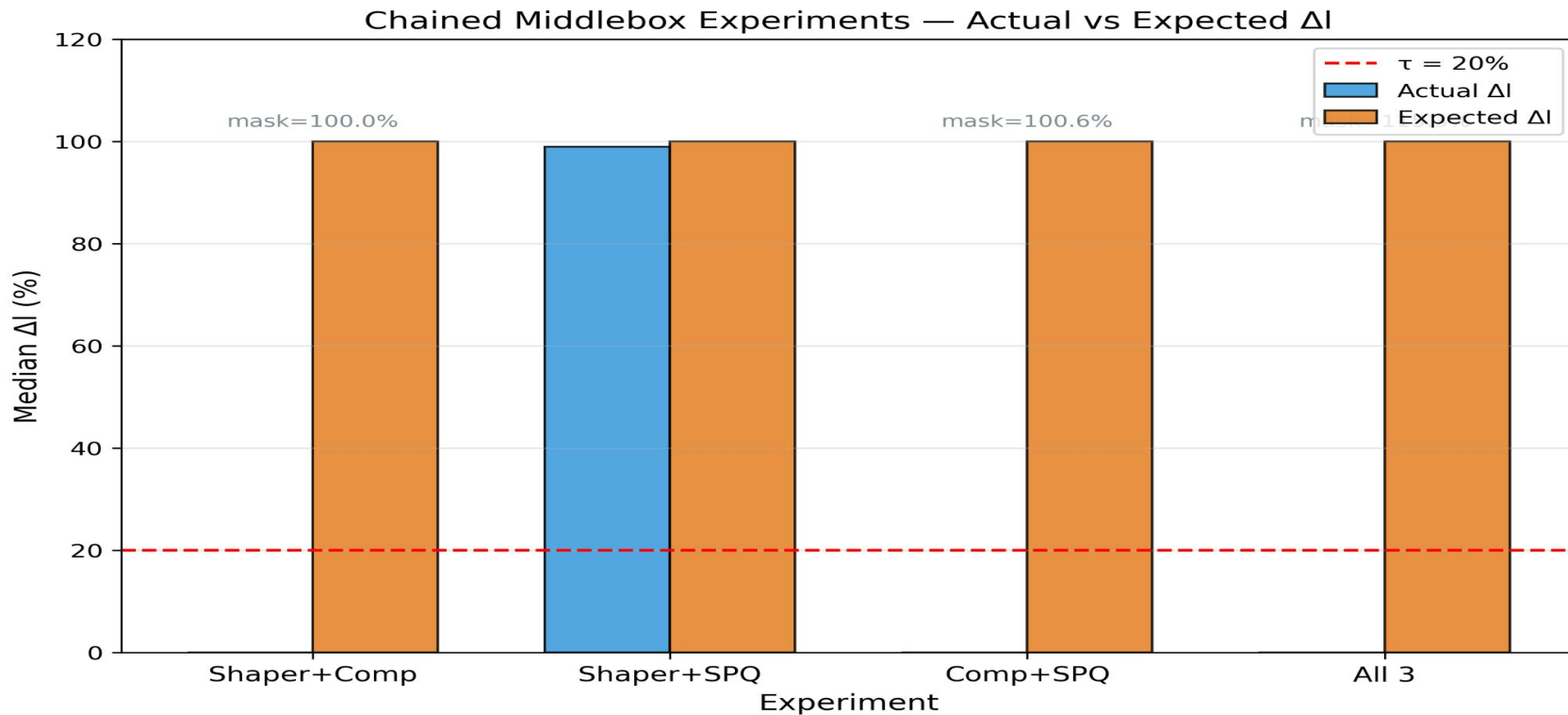
Novel contribution — first empirical chaining study

Chain	ΔI Actual	ΔI Expected	Masking M	Finding
Shaper + Comp	0%	100%	100%	Complete masking
Shaper + SPQ	99%	100%	~0%	No masking
Comp + SPQ	0%	100%	100%	Complete masking
All 3	-12.7%	100%	>100%	Signal inverted

Masking coefficient: $M = 1 - (\Delta I_{\text{chain}} / \Delta I_{\text{expected}})$

Phase 2: Chaining — The Masking Effect

Chained Middle box experiments-Actual vs Expected ΔI



Why Masking Happens — Analysis

Three fundamentally different mechanisms:

EXP4 (Shaper + Compressor) — M=100%

Shaper shapes both trains equally (both port 9000) → both arrive at 2 Mbps → compressor's discrimination condition $r > cp \times \alpha PH$ no longer satisfied → compressor contributes zero additional loss

EXP6 (Comp + SPQ) — M=100%

All SPQ packets carry zeros → compressor classifies ALL as low entropy → rewrites ALL DSCP to 0 → SPQ sees no DSCP EF packets → band 0 always empty → no starvation

EXP7 (All 3) — M>100%, signal inverted

Shaper shapes LP separation packets in HP phase → hurts probe-B's helper packets → net $IB > IA$ → ΔI goes negative → framework reports "no MB" when 3 exist

Phase 3: TTL Localization Failure

Novel finding — TTL probing fails for all MB types

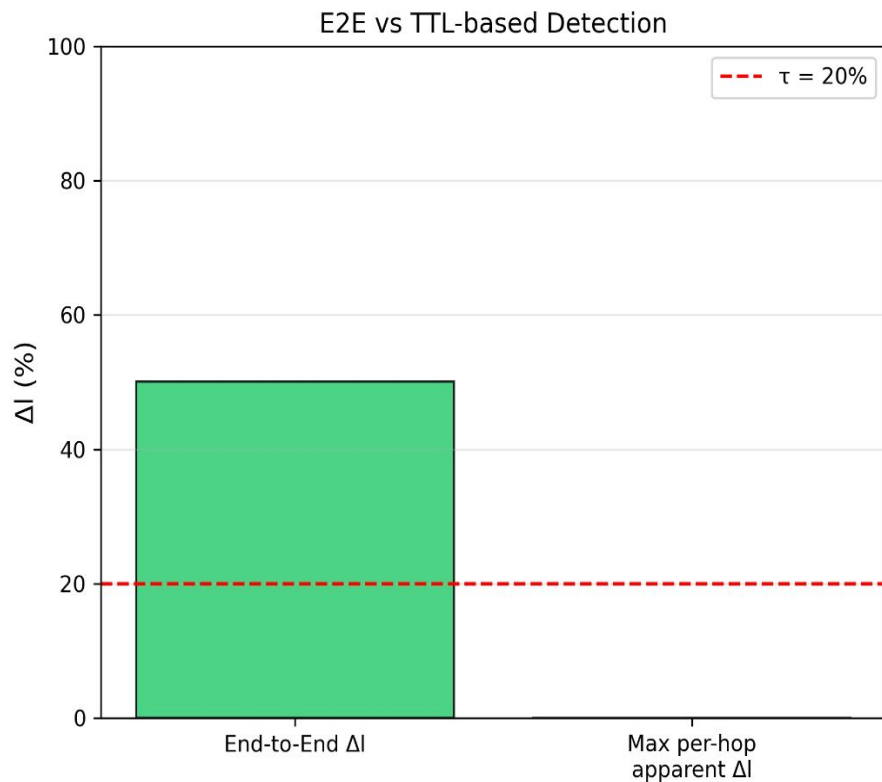
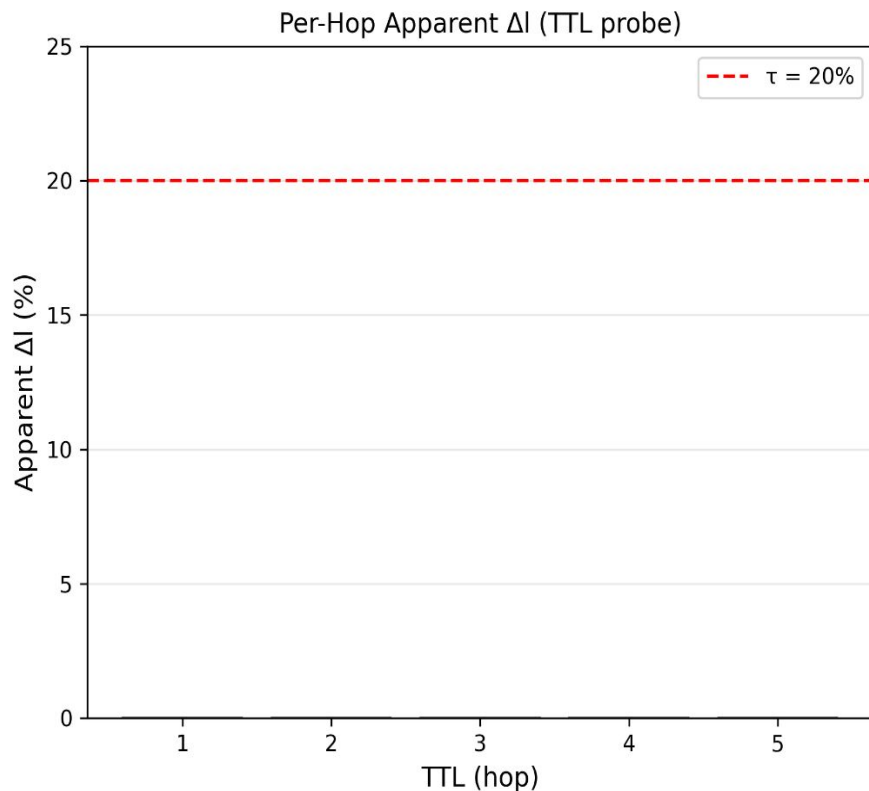
Method	Shaper	Compressor
End-to-end Δ	50.1% ✓ DETECTED	50.1% ✓ DETECTED
TTL apparent Δ at r_mb (hop 3)	-0.5% ✗ NOT DETECTED	-0.5% ✗ NOT DETECTED

Why TTL fails — architectural reason:

```
Packet arrives → TTL decrement → TTL=0? → ICMP generated (original packet)
                                   ↓ NO
                               Forwarding lookup → tc queue → shaper/compressor acts
```

TTL expiry happens BEFORE the forwarding plane. ICMP carries the original unmodified packet. The shaper and compressor never see TTL-expired packets.

Phase 3: TTL Localization Failure experimental Plot

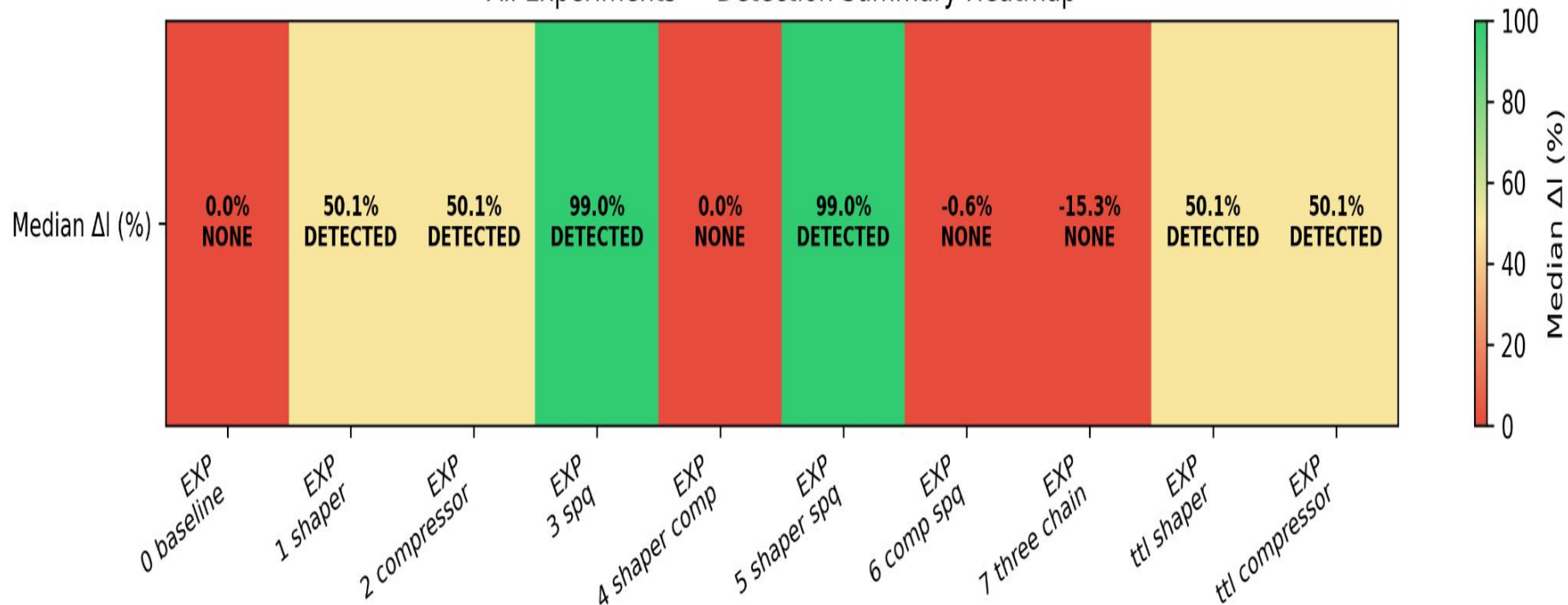


Summary of All Results — Complete Picture

Finding	Result
Single MB detection	All 3 detected (paper replicated) ✓
Same-class chaining	Complete masking ($\Delta I = 0\%$)
Different-class chaining	No masking ($\Delta I = 99\%$)
3-chain	Signal inversion ($\Delta I < 0\%$)
TTL localization	Fails for all behavioral MB types

Summary of All Results — Complete Picture

All Experiments — Detection Summary Heatmap



Conclusions and Future Work

What we proved:

- Paper's single-MB detection framework works — replicated with standard Linux tc
- Chaining is NOT simply additive — masking depends on classification signal overlap
- Same classification signals → masking. Different signals → reinforcement
- 3-chain can invert the detection signal — framework reports clean path when 3 MBs present
- TTL probing cannot localize behavioral MBs in Linux kernel — architectural limitation

Future Work:

- Rate-sweep probing to count middleboxes in a chain
- Entropy fingerprinting to identify MB type from ΔI response curve
- Larger topologies with more normal routers between MBs
- Dynamic evasion (paper Section VIII)

References

- Pournaghshband & Reiher, "End-to-End Detection of Middlebox Interference," IEEE NOMS 2024
- Chang, Rahimi & Pournaghshband, "Differentiated Service Queuing Disciplines in NS-3," SIMUL 2015
- Nguyen & Roughan, "Rigorous Statistical Analysis of Internet Loss Measurements," IEEE/ACM ToN 2013
- Detal et al., "Revealing Middlebox Interference with Tracebox," IMC 2013

Thank You